



Chapter 8

Digital Filter Structures



Main Contents in this chapter



1

Block Diagram Representation of the LTI Digital Filter

2

Basic FIR Filter Structure

3

Basic IIR Filter Structure



Block Diagram Representation of Digital Filter

- *Basic Building Blocks*
- *Delay-Free Loop Problem*
- *Canonic & Equivalent Structure*



Introduction



- The actual implementation of an LTI digital filter could be either in **software or hardware** form, depending on applications.
- The **structural representation** provides the relations between some pertinent internal variables with the input and the output that in turn provide the keys to the implementation.
- In this lecture, we will focus on the **block diagram representation** of FIR and IIR digital filters

Introduction



But, how can we realize those filters?

For a LTI FIR system we know

$$y[n] = \sum_{k=N_1}^{N_2} h[k]x[n-k]$$

For a LTI IIR system

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \quad (8.1)$$

Digital filter Structures



- The convolution sum description of an LTI discrete-time system can, in principle, be used to implement the system
- For an IIR finite-dimensional system this approach is not practical as here the impulse response is of **infinite length**

However, a direct implementation of the FIR finite-dimensional system is practical

Introduction



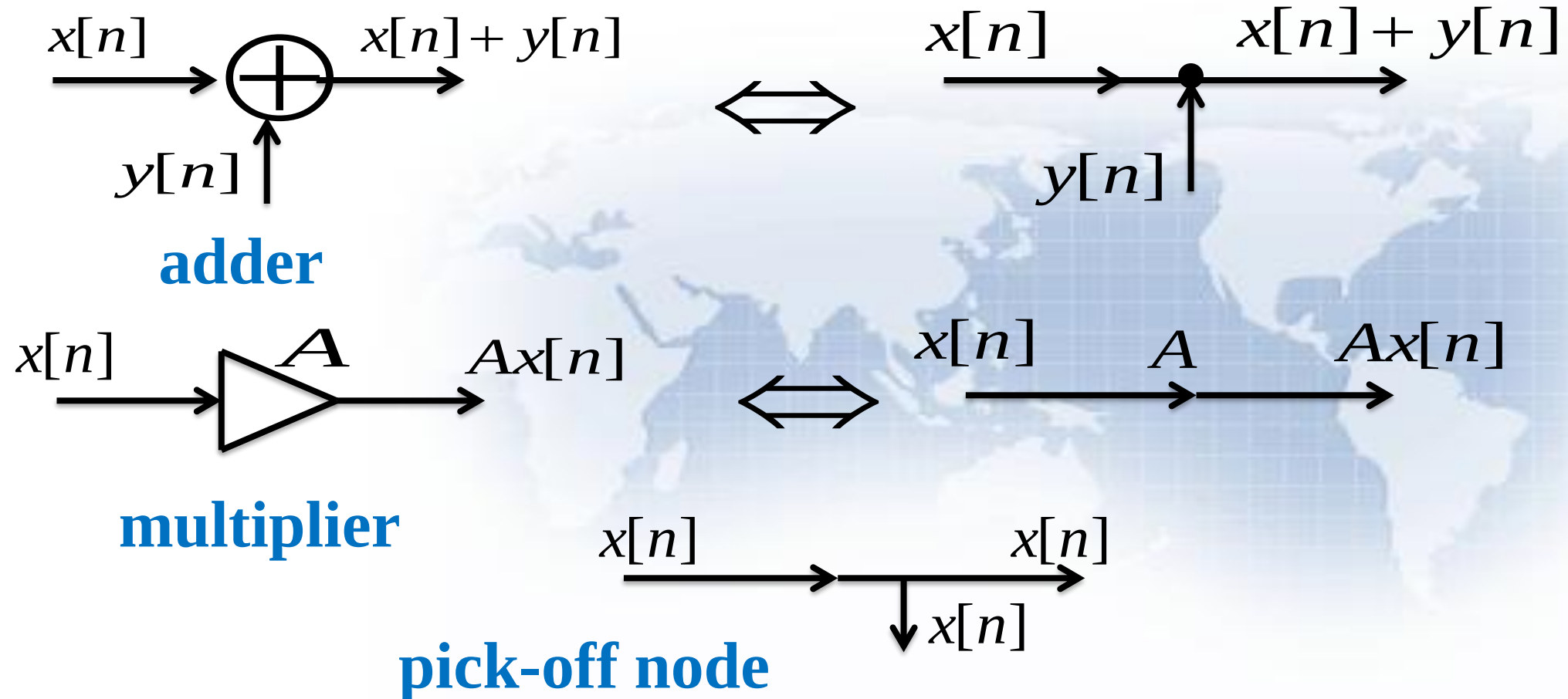
The above representations are **not clear enough** for us to implement those filters(such as in hardware form or improving our algorithms).

The structural representation provides the **relations between** some pertinent internal variables with the input and the output that in turn provide the **keys** to the implementation.

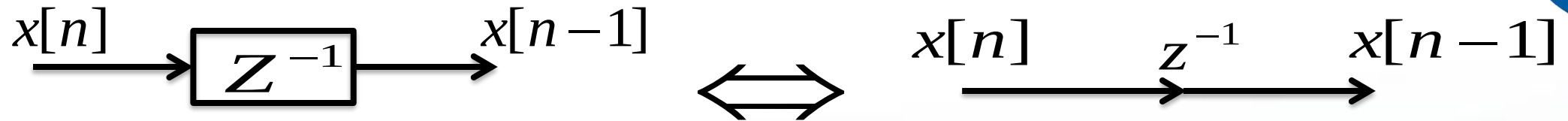
Basic Building Blocks



- The basic building blocks and signal-flowing diagram



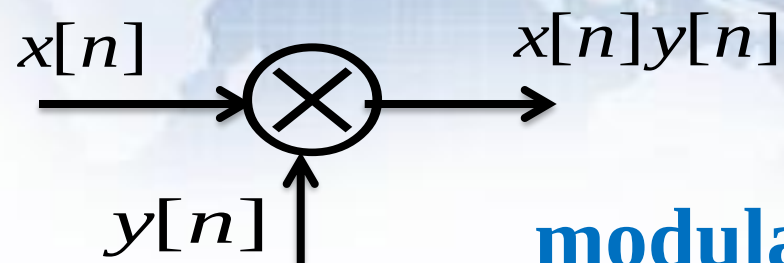
Basic Building Blocks



unit delay



unit advance



modulator

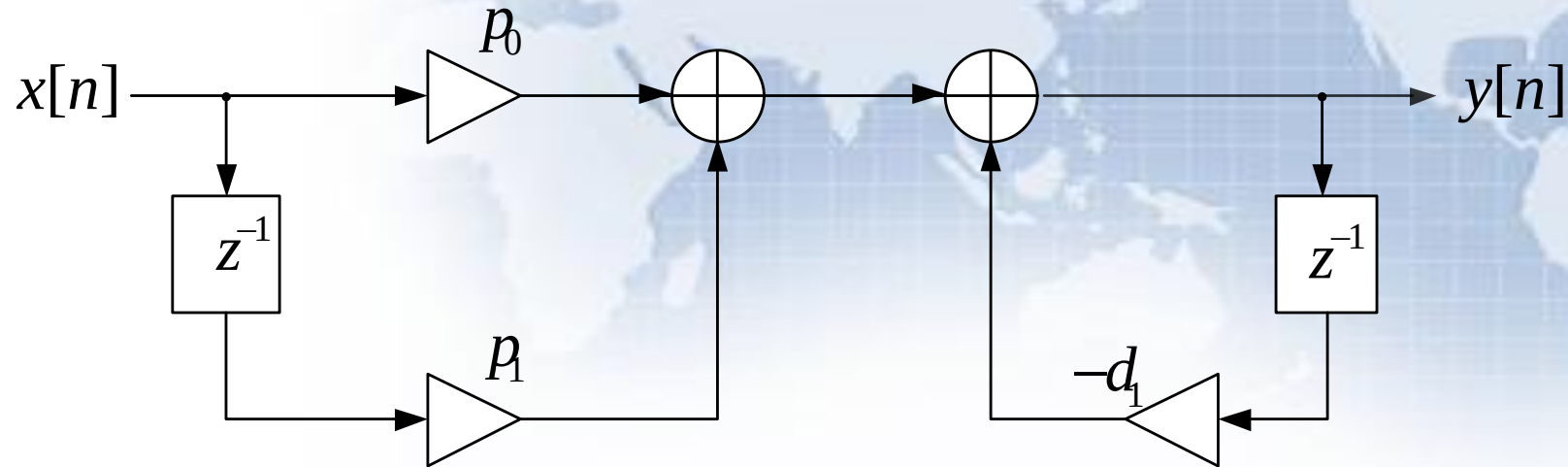
Basic Building Blocks



- **Example:**

$$y[n] = -d_1 y[n-1] + p_0 x[n] + p_1 x[n-1]$$

- **Its block diagram is as below:**



Block Diagram Representation

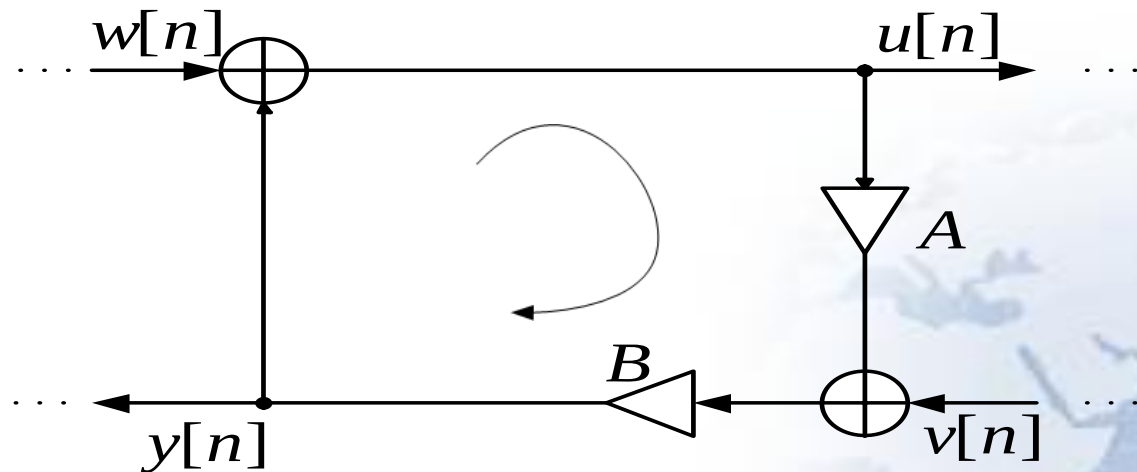


- The advantages of block diagram representation:
 - Easy to write down the computational algorithm
 - Easy to determination the relation between the input and output
 - Easy to derive other “equivalent” block diagram yielding different algorithms
 - Easy to determine the hardware requirements

Delay-Free Loop



- The Delay-Free Loop Problem**



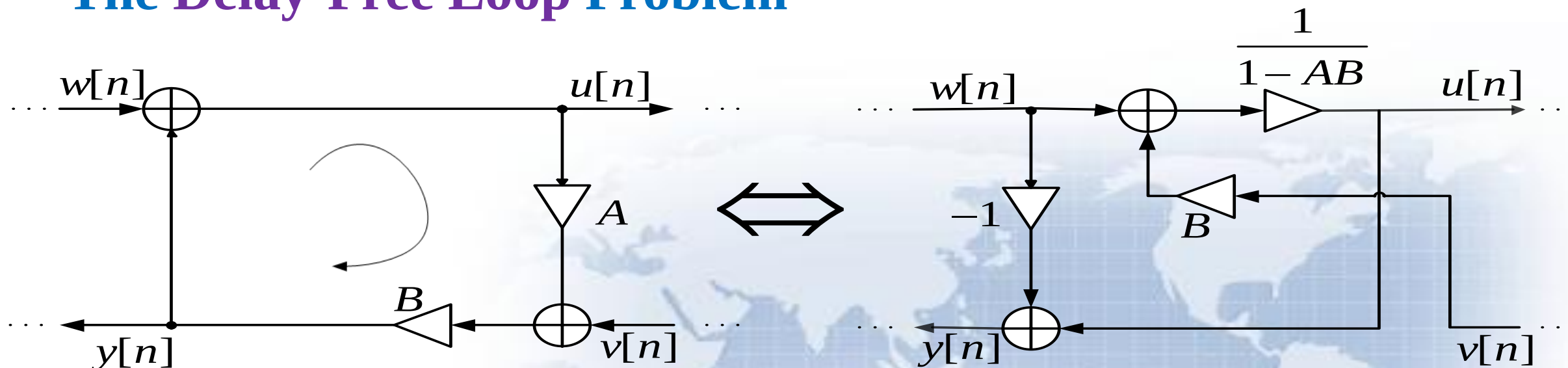
$$y[n] = B \{ A(w[n] + y[n]) + v[n] \}$$

Output $y[n]$ depends on $y[n]$ itself. Physically impossible !

Delay-Free Loop



- The Delay-Free Loop Problem



$$y[n] = B \{ A(w[n] + y[n]) + v[n] \} \Leftrightarrow y[n] = \frac{AB}{1 - AB} w[n] + \frac{B}{1 - AB} v[n]$$

Delay-free loop

No delay-free loop

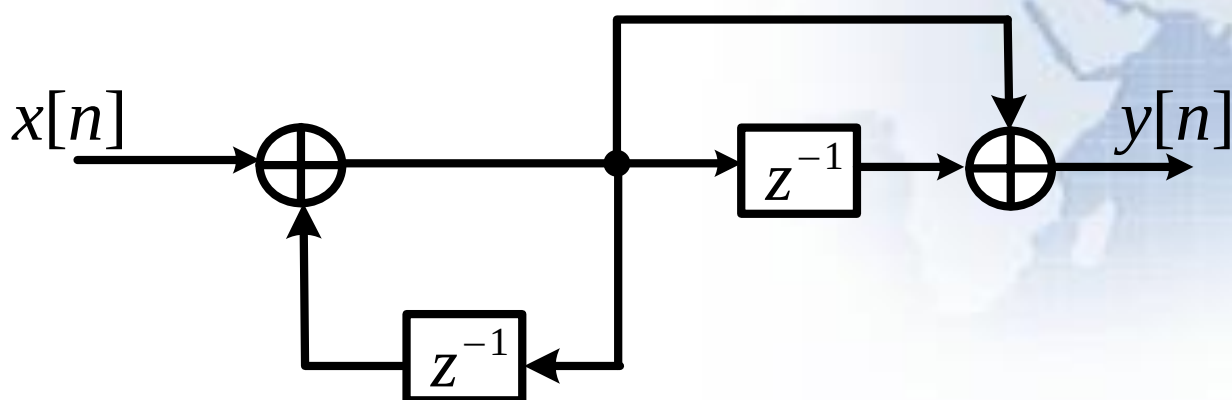
Canonic & Equivalent Structure



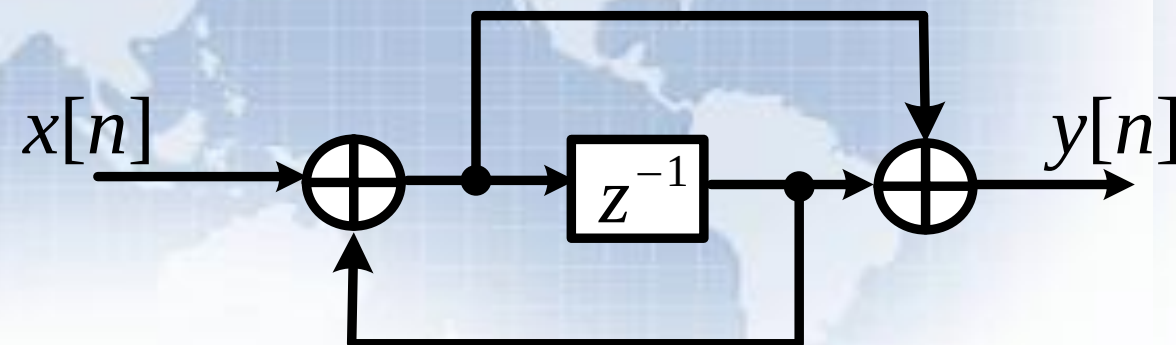
- *Canonic structure*

- *The number of delays = the order of the difference equation*

- *Example* $y[n] = y[n - 1] + x[n] + x[n - 1]$



Non-canonic structure

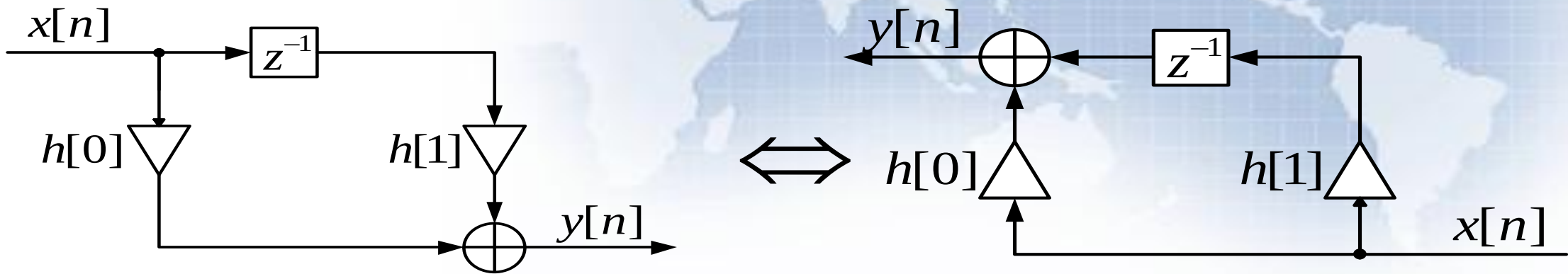


Canonic structure

Canonic & Equivalent Structure



- **Equivalent structure via transpose operation**
 - (1) Reverse all paths
 - (2) Replace pick-off nodes by adders, and vice versa
 - (3) Interchange the input and the output nodes





Basic FIR & IIR Digital Filter Structure

- *FIR Filter Structure*
- *IIR Filter Structure*



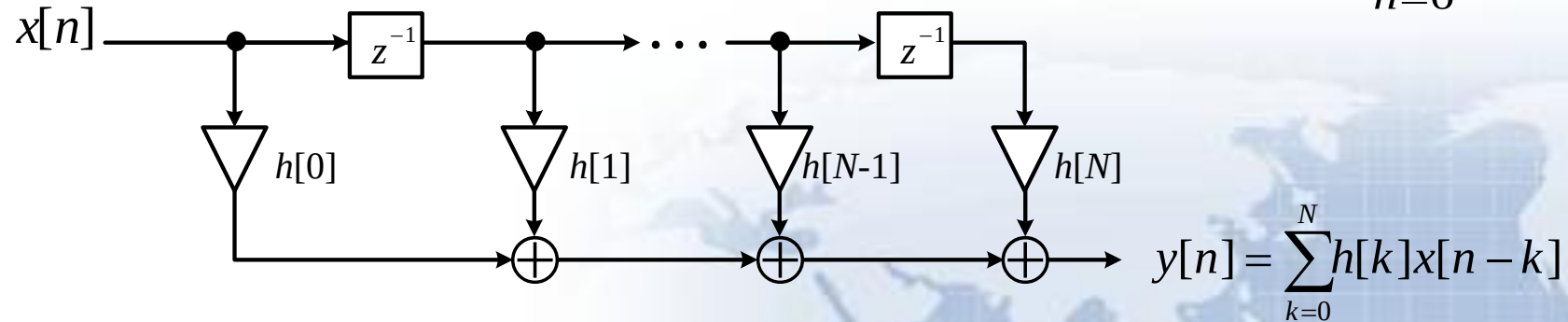
Basic FIR Filter Structure



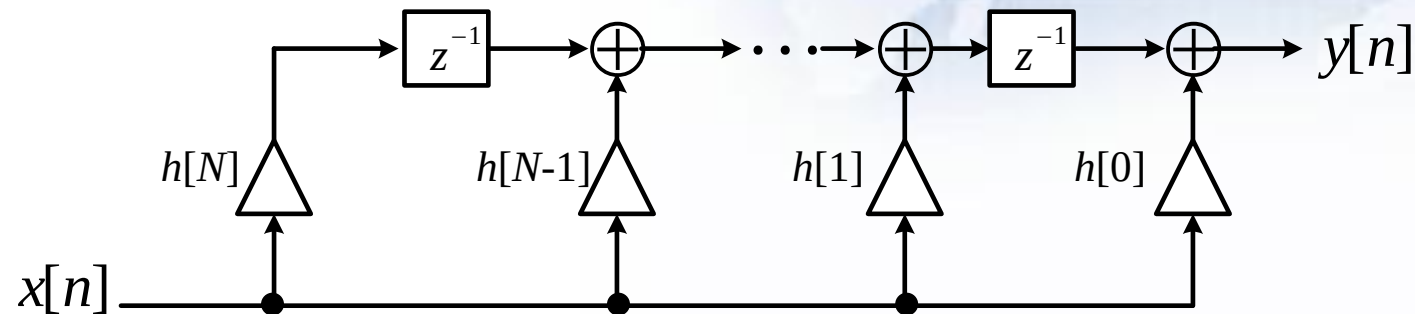
- Direct Form

➤ A causal FIR filter of order N

$$H(z) = \sum_{n=0}^N h[n]z^{-n}$$



and its transpose



Direct form is
canonic structure

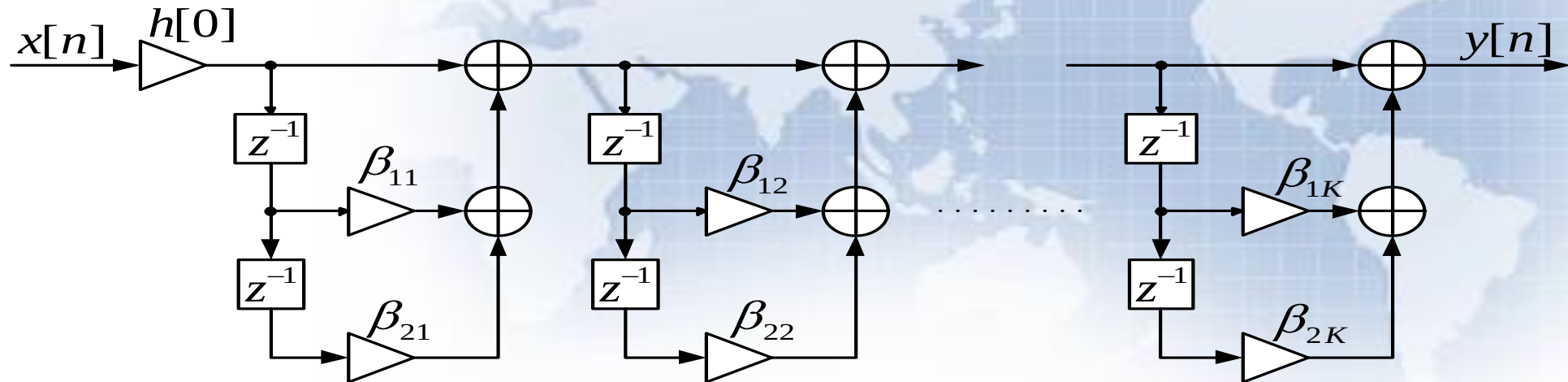
Basic FIR Filter Structure



- Cascade Form

$$H(z) = h[0] \prod_{k=1}^K \left(1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2} \right)$$

where $H(z)$ is divided into 1st-order (i.e., $\beta_{2k} = 0$) or 2nd-order (i.e., $\beta_{2k} \neq 0$) transfer functions.



Cascade form is also canonic structure

8.3 Basic FIR Filter Structure



8.3.3 Polyphase realization

$$\begin{aligned} H(z) &= h[0] + h[1]z^{-1} + h[2]z^{-2} + h[3]z^{-3} + h[4]z^{-4} \\ &\quad + h[5]z^{-5} + h[6]z^{-6} + h[7]z^{-7} + h[8]z^{-8} \\ &= (h[0] + h[2]z^{-2} + h[4]z^{-4} + h[6]z^{-6} + h[8]z^{-8}) \\ &\quad + z^{-1}(h[1] + h[3]z^{-2} + h[5]z^{-4} + h[7]z^{-6}) \\ &= E_0(z^2) + z^{-1}E_1(z^2) \quad \text{See Fig8.7c} \\ &= E_0(z^3) + z^{-1}E_1(z^3) + z^{-2}E_2(z^3) \quad \text{See Fig8.7b} \\ &= E_0(z^4) + z^{-1}E_1(z^4) + z^{-2}E_2(z^4) + z^{-3}E_3(z^4) \quad \text{See Fig8.7a} \end{aligned}$$

8.3 Basic FIR Filter Structure



The above structures are noncanonic. For

$$H(z) = \sum_{m=0}^{L-1} z^{-m} E_m(z^L) \quad (8.17)$$

Where

$$E_m(z) = \sum_{n=0}^{\lfloor (N+1)/L \rfloor} h[Ln + m] z^{-n}, \quad 0 \leq m \leq L-1, \quad (8.18)$$

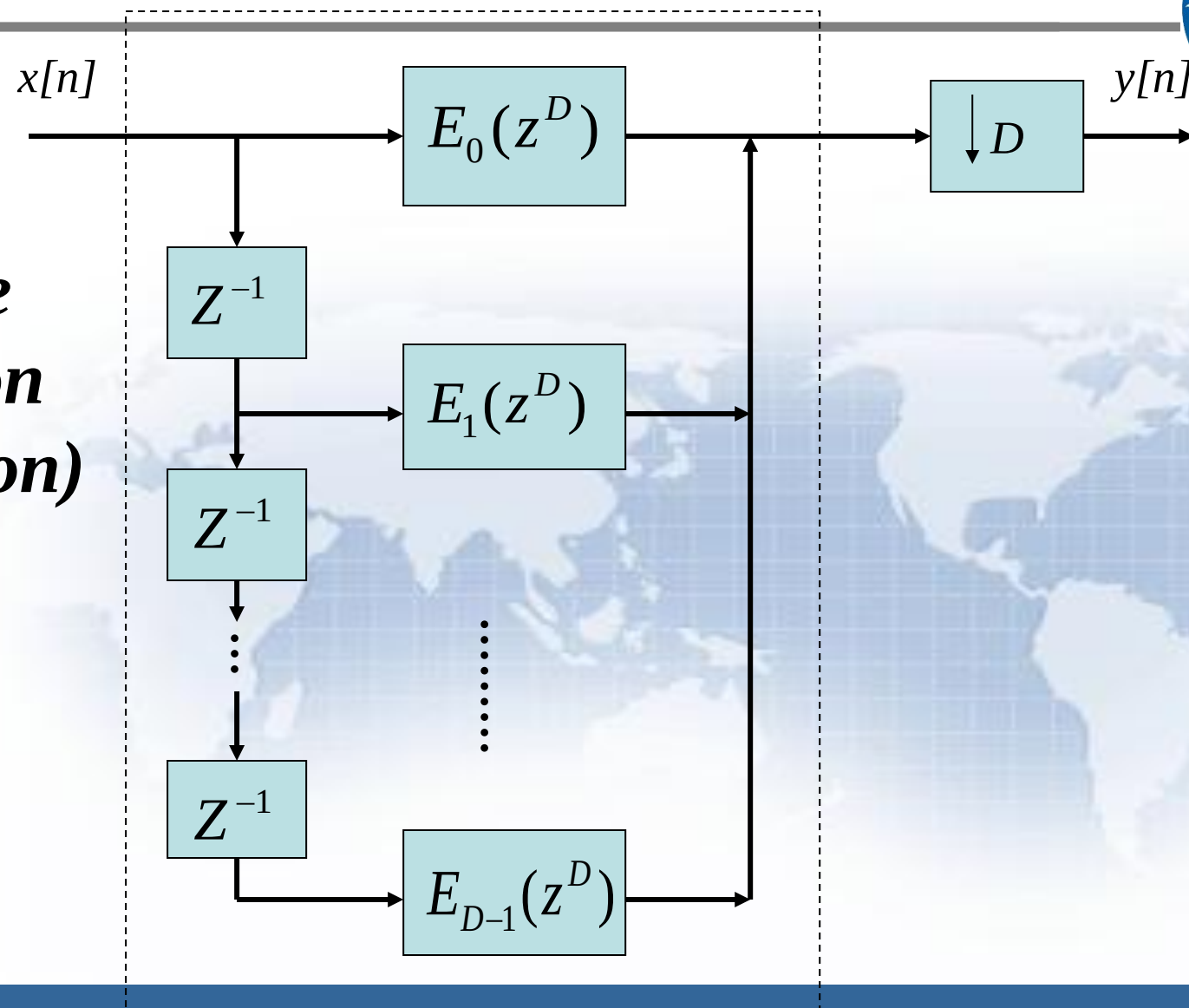
If the delay(z^{-L}) in all subfilters are shared, the structure is canonic. See Fig8.8

The polyphase structures are often used in **multirate digital signal processing** applications for computation efficient realizations.

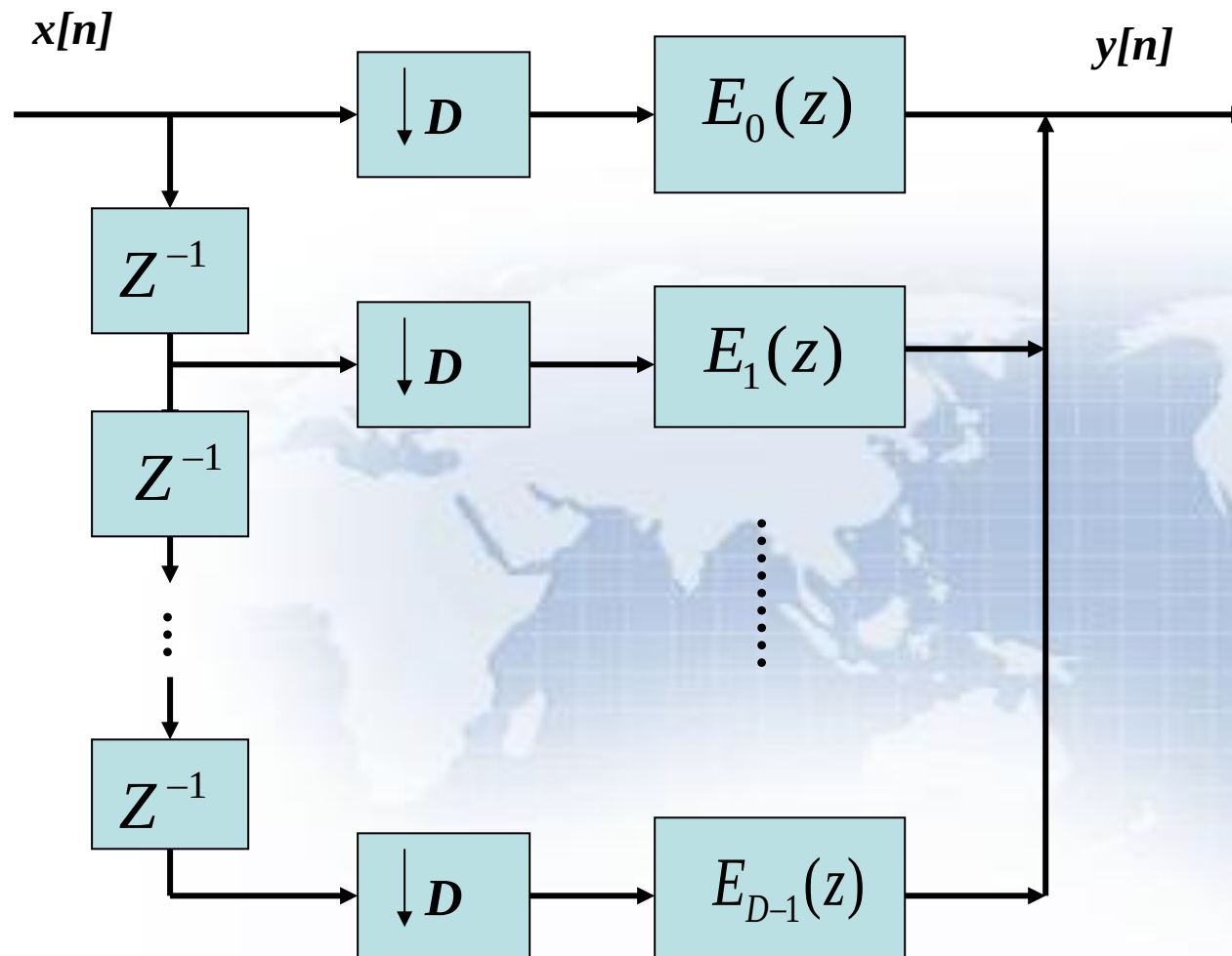
Application:



Polyphase Realization (decimation)



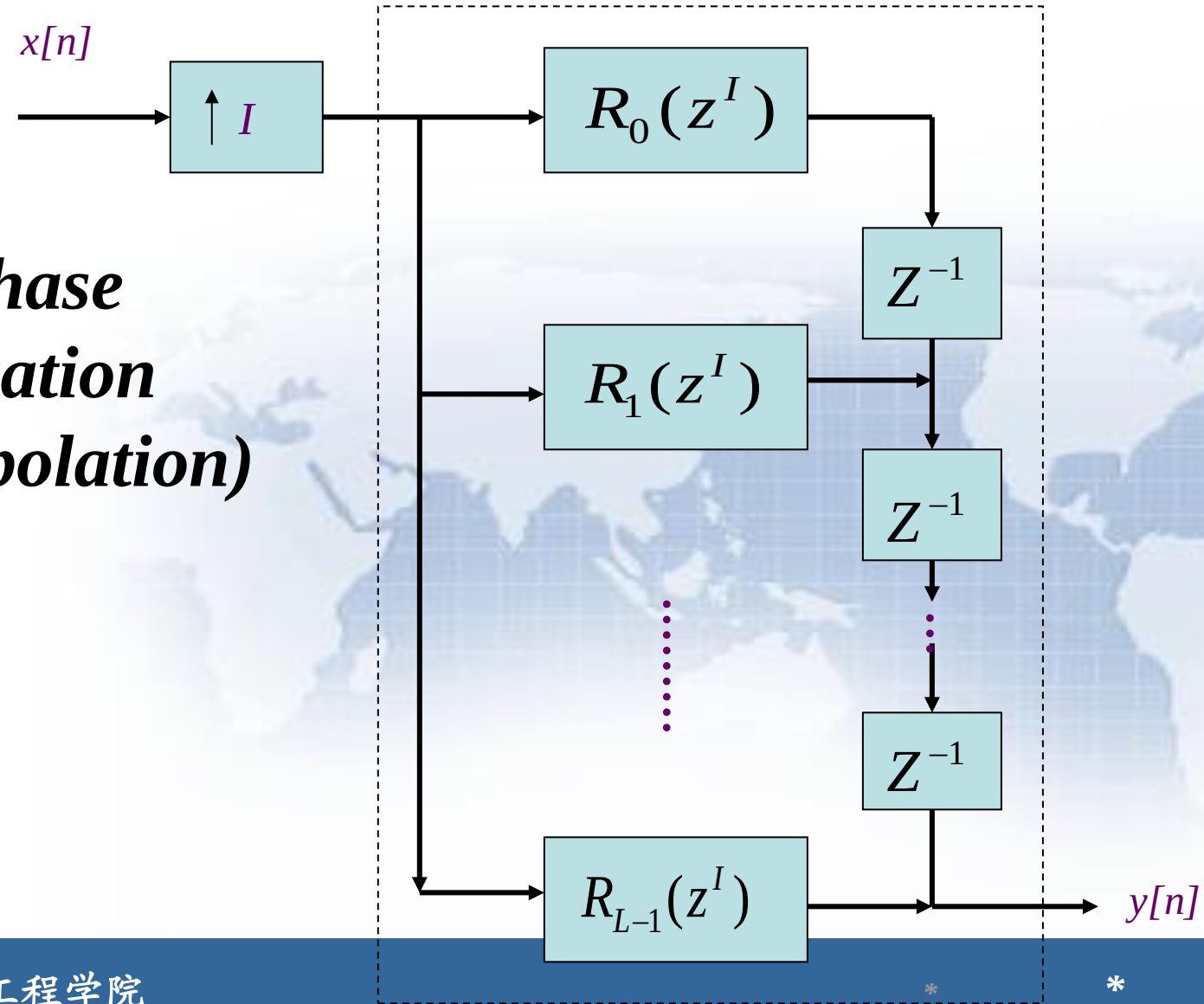
Application:



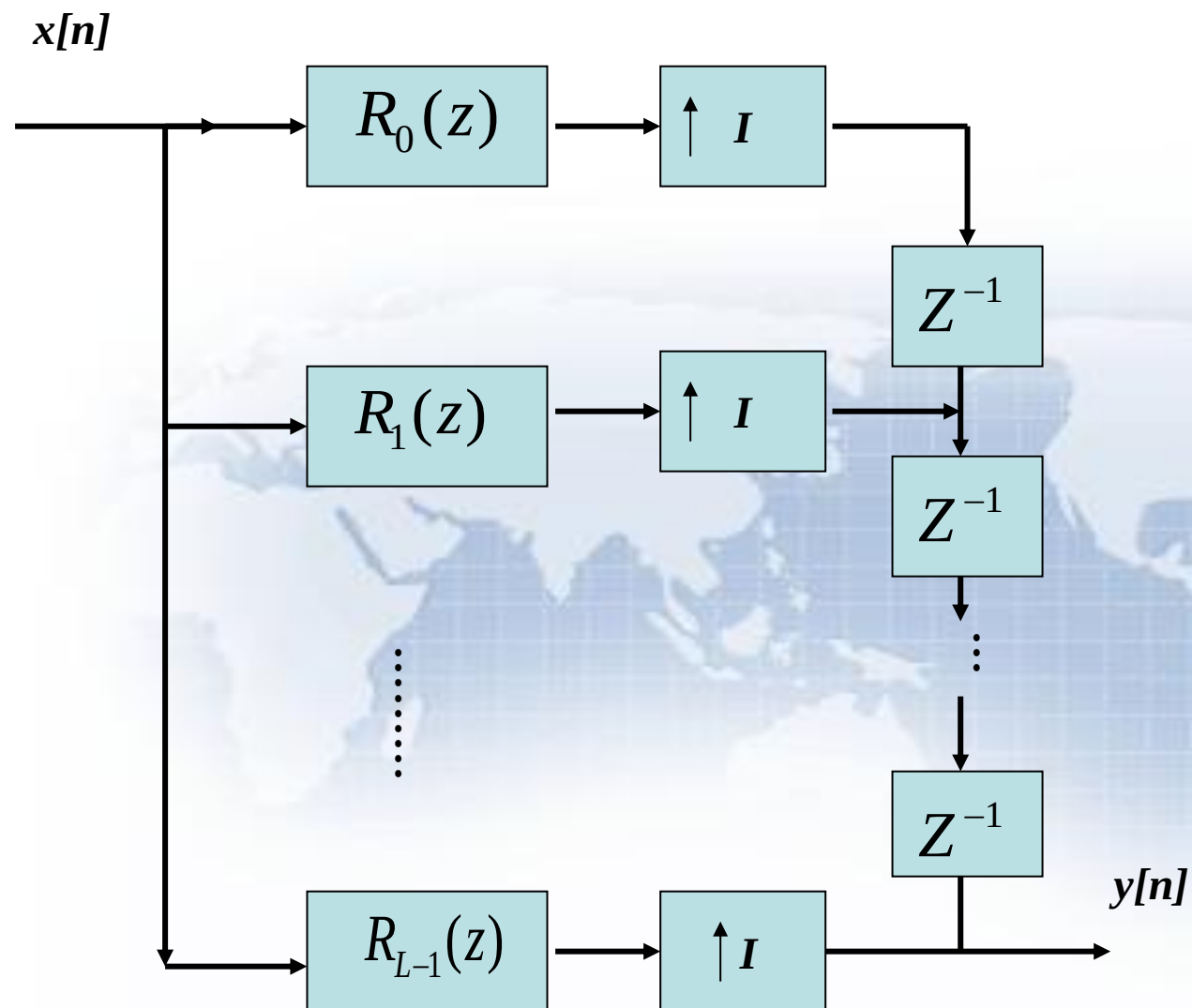
Application:



Polyphase Realization (interpolation)



Application:



8.3 Basic FIR Filter Structure



8.3.4 Linear-Phase FIR Structures

Linear-Phase FIR filter : $h[n]$ is symmetric, or antisymmetric:

$$h[n] = h[N - n] \text{ or } h[n] = -h[N - n]$$

The characterization can reduce the number of multipliers (almost a half) indirect form.

Example

The implementations of length 7 and 8 with symmetric impulse response :

8.3 Basic FIR Filter Structure



$$H(z) = h[0](1 + z^{-6}) + h[1](z^{-1} + z^{-5}) \\ + h[2](z^{-2} + z^{-4}) + h[3]z^{-3}$$

$$H(z) = h[0](1 + z^{-7}) + h[1](z^{-1} + z^{-6}) \\ + h[2](z^{-2} + z^{-5}) + h[3](z^{-3} + z^{-4})$$

It is noted that a similar savings occurs in the case of an FIR filter with an antisymmetric impulse response.

8.3 Basic FIR Filter Structure

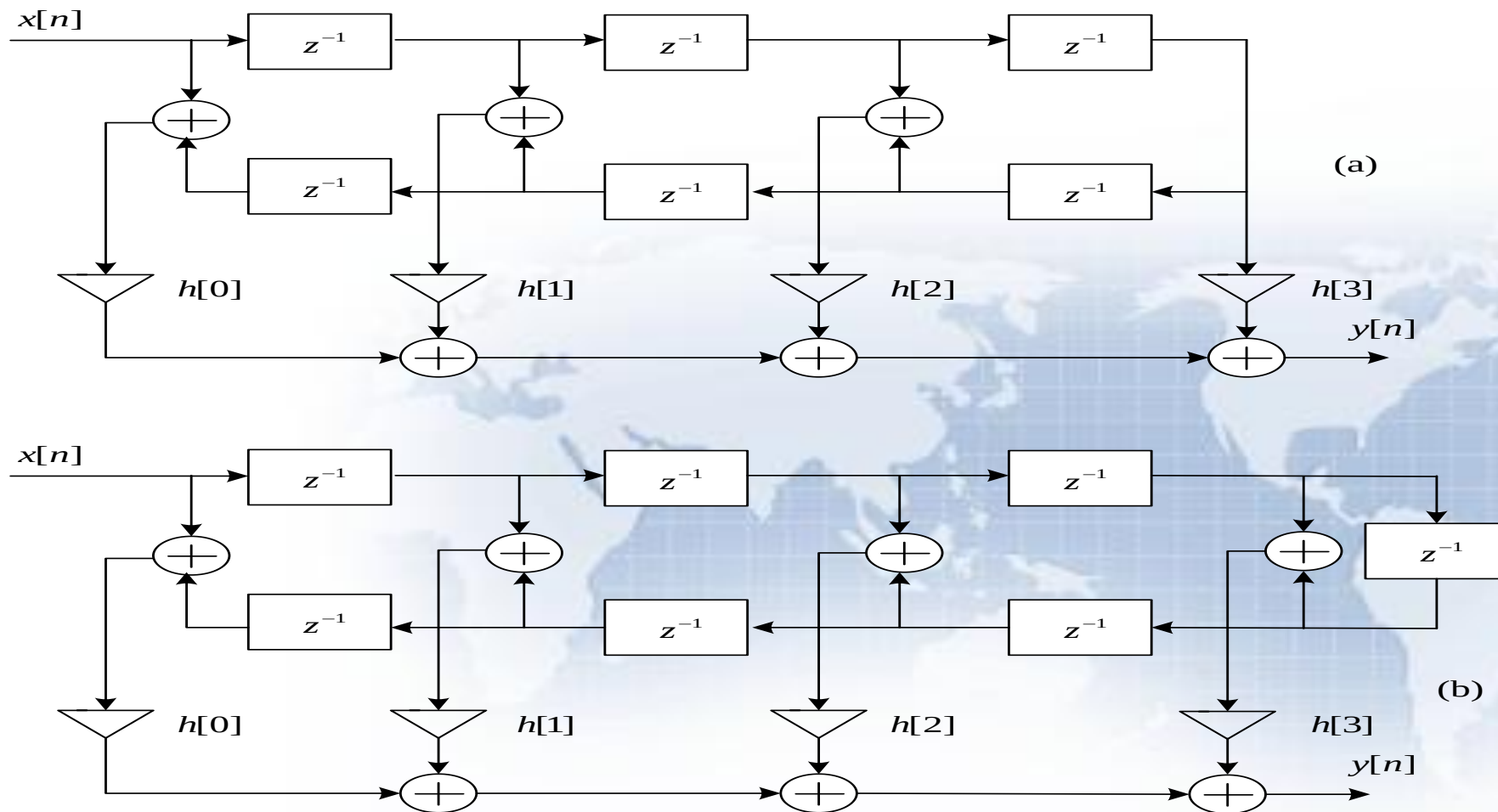


Fig 8.9 Linear-phase FIR structures:(a)Type 1,(b)Type 2

8.3 Basic FIR Filter Structure



Example: Given an FIR filter as below

$$H(z) = 1 - 2.05z^{-1} + 3.2025z^{-2} - 1.05z^{-3} - 1.05z^{-4} \\ + 3.2025z^{-5} - 2.05z^{-6} + z^{-7}$$

Its cascade form structure is wanted.

Solution: $\because N = 7 \quad \therefore 1\text{-order zero } z_1 = 1 \text{ or } z_1 = -1$

because the coefficient of z^{-7} is positive

$$\therefore z_1 = -1, \quad H_1(z) = 1 + z^{-1}$$

$$H'(z) = H(z) / H_1(z) = 1 - 3.05z^{-1} + 6.2525z^{-2} \\ - 7.3025z^{-3} - 6.2525z^{-4} + 3.05z^{-5} - z^{-6}$$

8.3 Basic FIR Filter Structure



$$\text{Set } H_2(z) = 1 + az^{-1} + z^{-2},$$

$$H_3(z) = 1 + bz^{-1} + cz^{-2} + bz^{-3} + z^{-4}$$

$$\therefore H'(z) = H_2(z)H_3(z)$$

$$\left. \begin{array}{l} a + b = -3.05 \\ 1 + ab + c = 6.2525 \\ 2b + ac = 7.3025 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} a = -1 \\ b = -2.05 \\ c = 3.2025 \end{array} \right.$$

$$\therefore H_2(z) = 1 - z^{-1} + z^{-2},$$

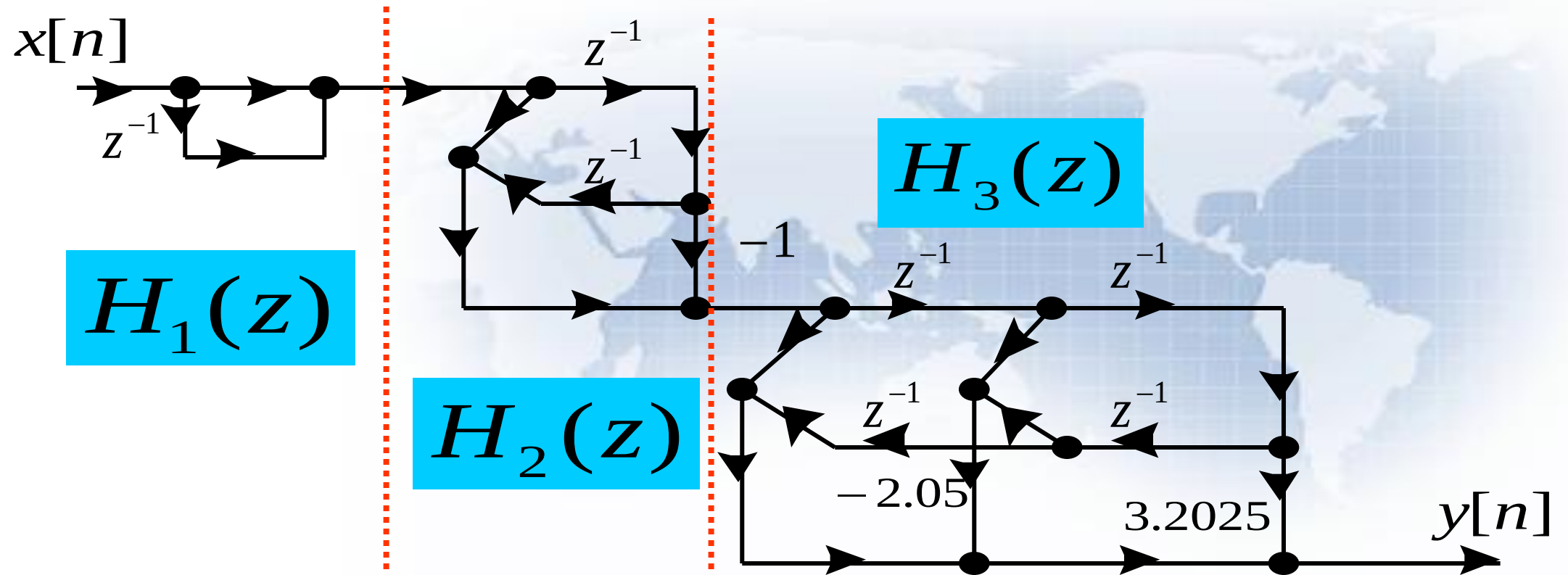
$$H_3(z) = 1 - 2.05z^{-1} + 3.2025z^{-2} - 2.05z^{-3} + z^{-4}$$

8.3 Basic FIR Filter Structure



$$\because H(z) = H_1(z)H_2(z)H_3(z)$$

Its cascade form structure is as below:



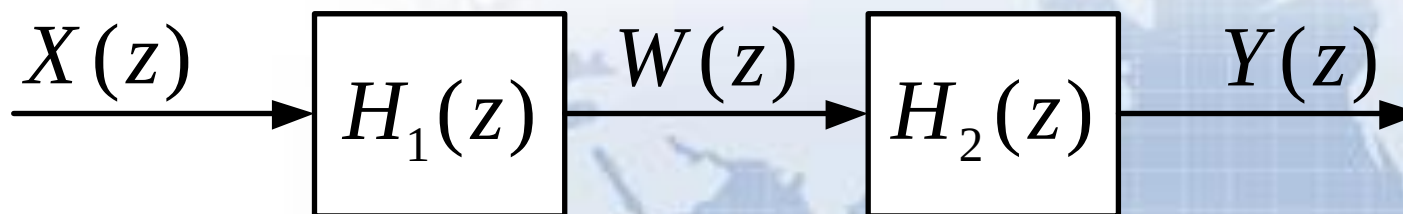
Basic IIR Filter Structure



- **Direct Form**

$$H(z) = \frac{P(z)}{D(z)} = \frac{p_0 + p_1z^{-1} + p_2z^{-2} + p_3z^{-3}}{1 + d_1z^{-1} + d_2z^{-2} + d_3z^{-3}}$$

➤ **A possible realization scheme**



$$H_1(z) = \frac{W(z)}{X(z)} = p_0 + p_1z^{-1} + p_2z^{-2} + p_3z^{-3}$$

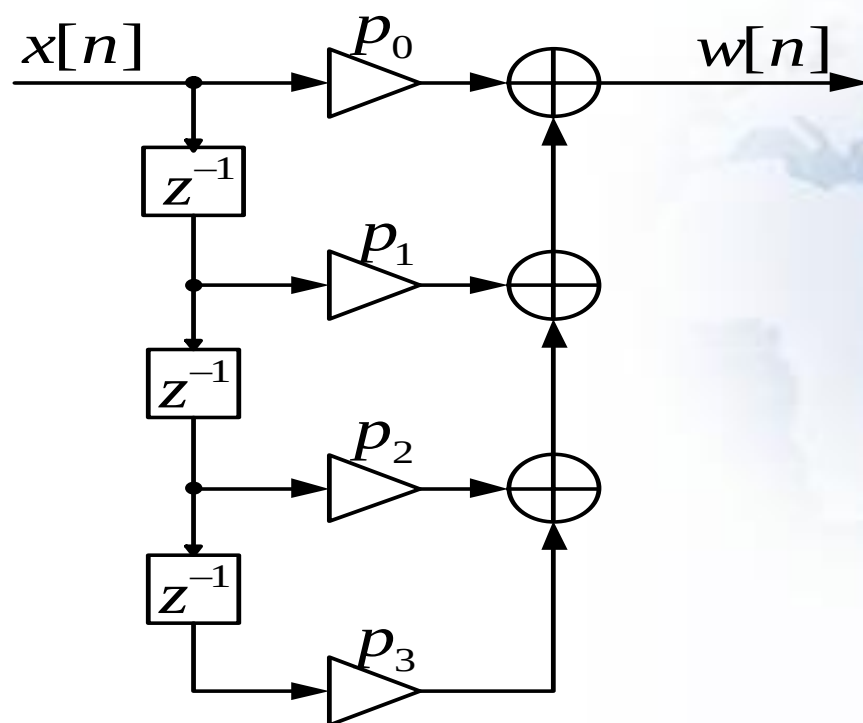
$$H_2(z) = \frac{Y(z)}{W(z)} = \frac{1}{1 + d_1z^{-1} + d_2z^{-2} + d_3z^{-3}}$$

Basic IIR Filter Structure



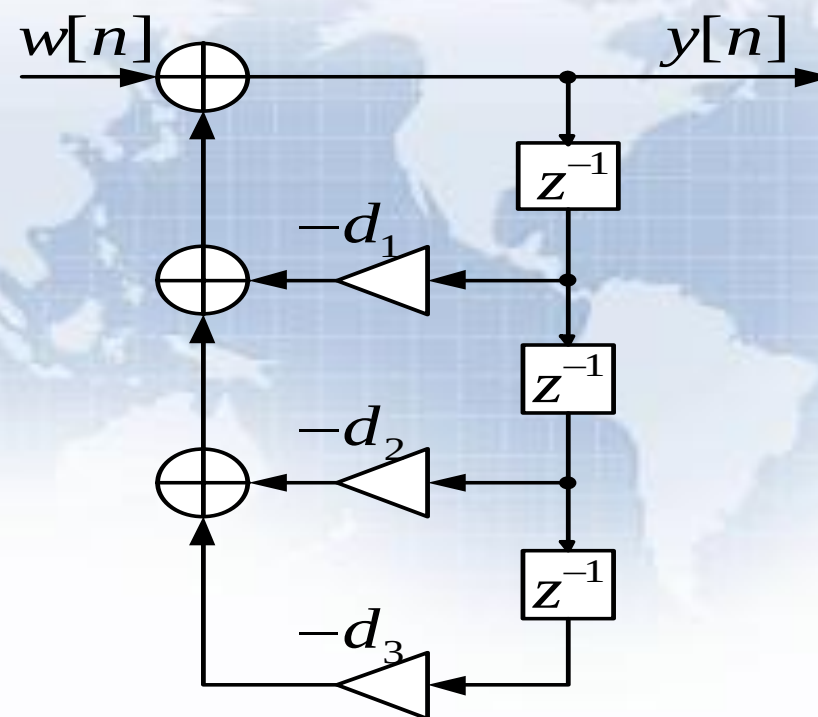
$$H_1(z) = p_0 + p_1z^{-1} + p_2z^{-2} + p_3z^{-3}$$

$$w[n] = \sum_{k=0}^3 p_k x[n-k]$$

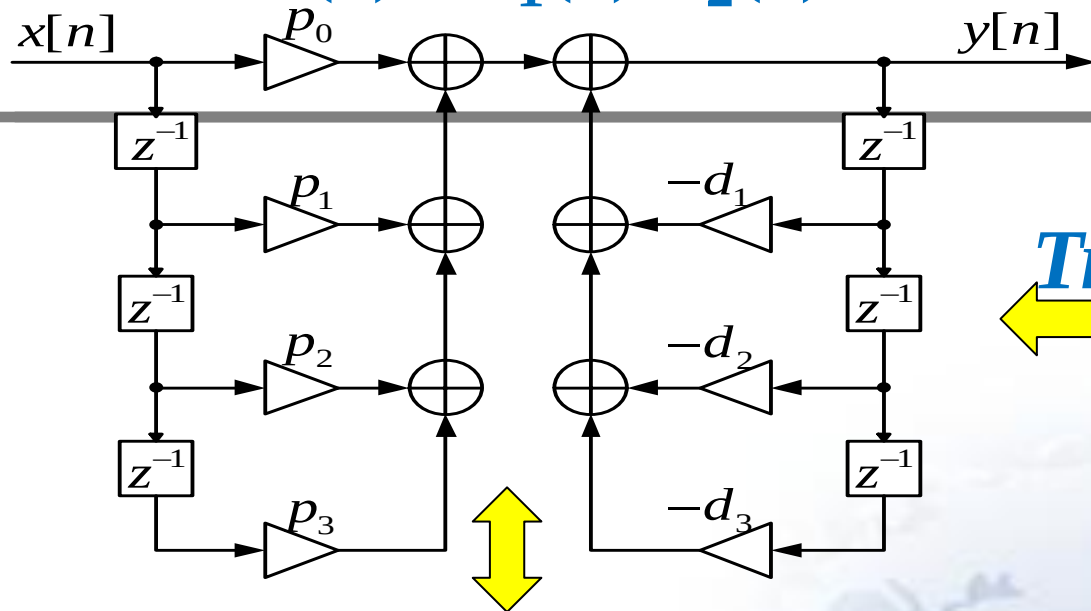


$$H_2(z) = \frac{1}{1 + d_1z^{-1} + d_2z^{-2} + d_3z^{-3}}$$

$$y[n] = w[n] - \sum_{k=1}^3 d_k y[n-k]$$

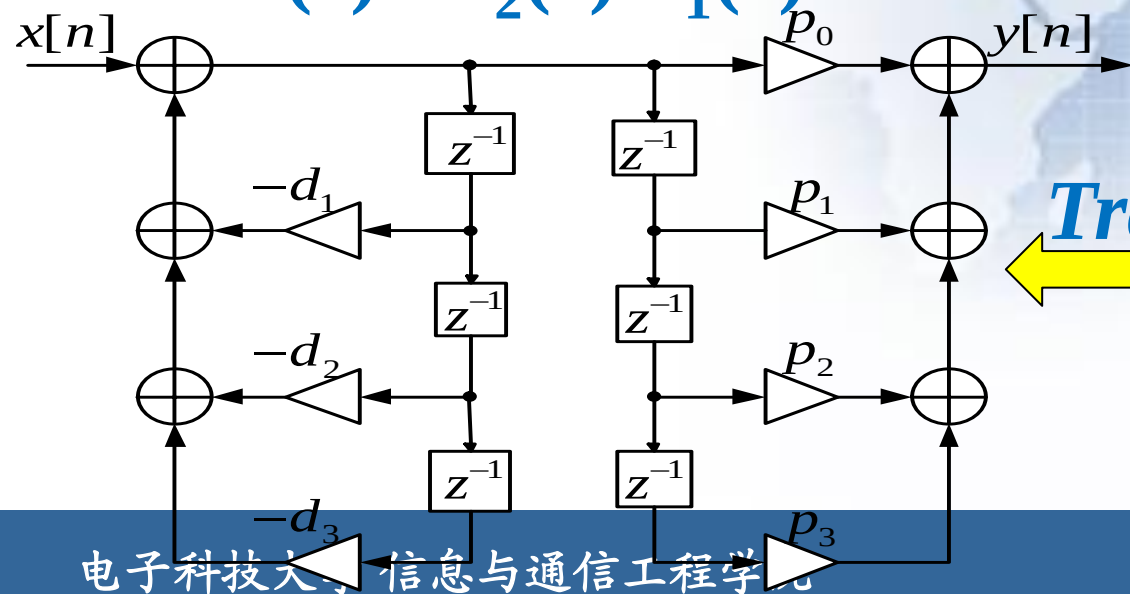


$$H(z) = H_1(z)H_2(z)$$

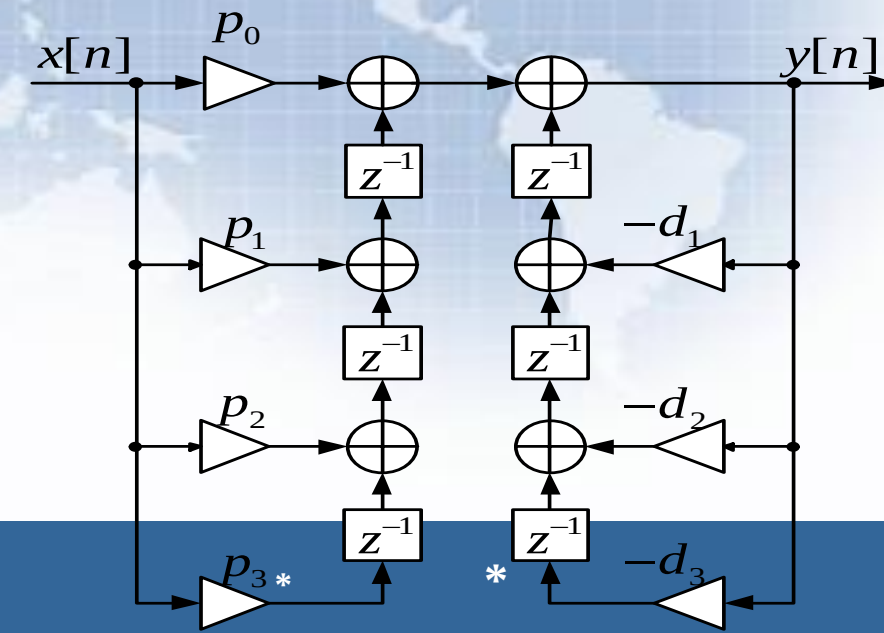
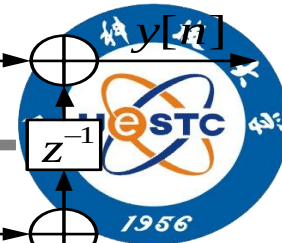
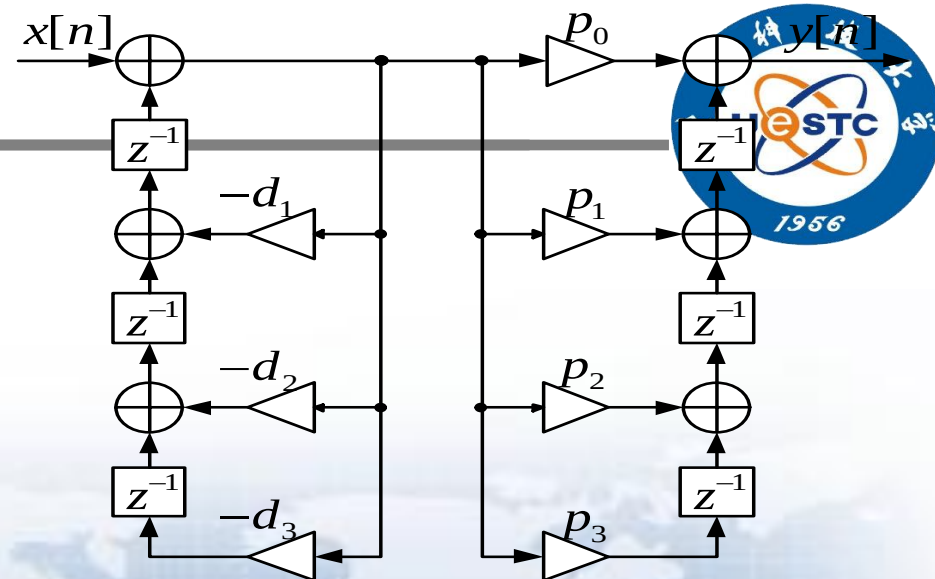


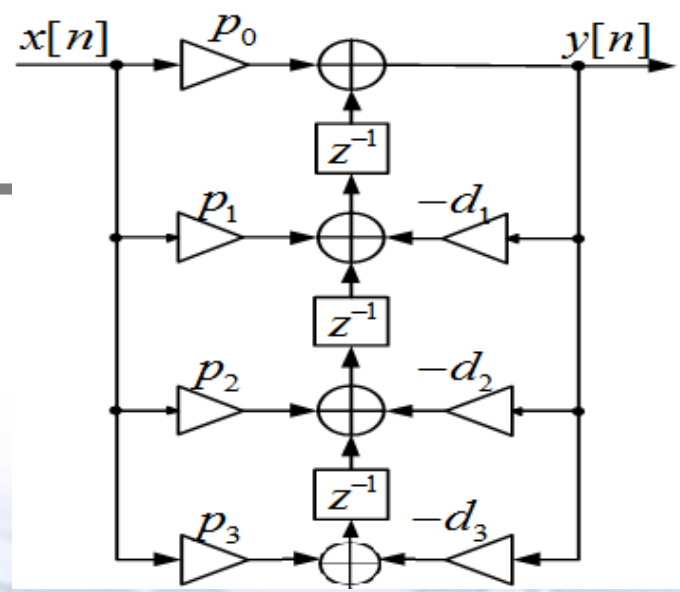
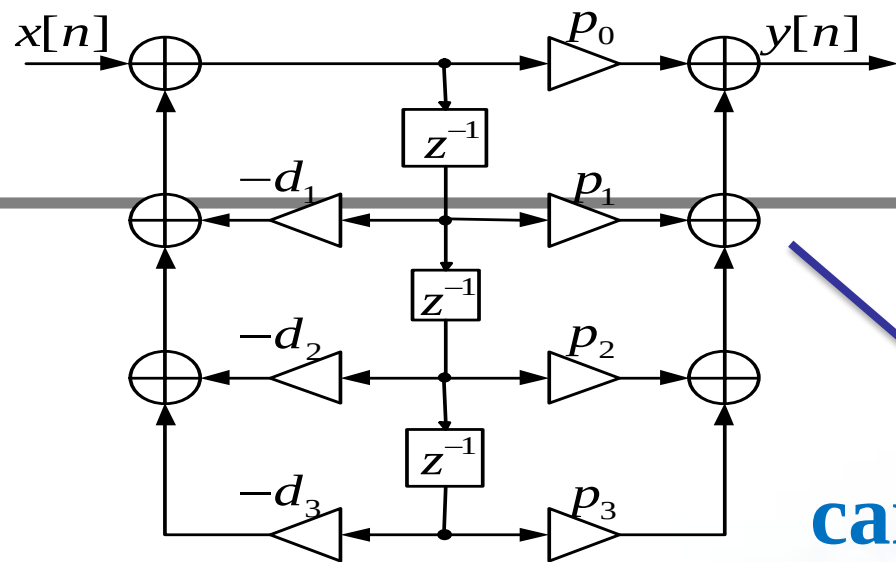
Transpose

$$H(z) = H_2(z)H_1(z)$$



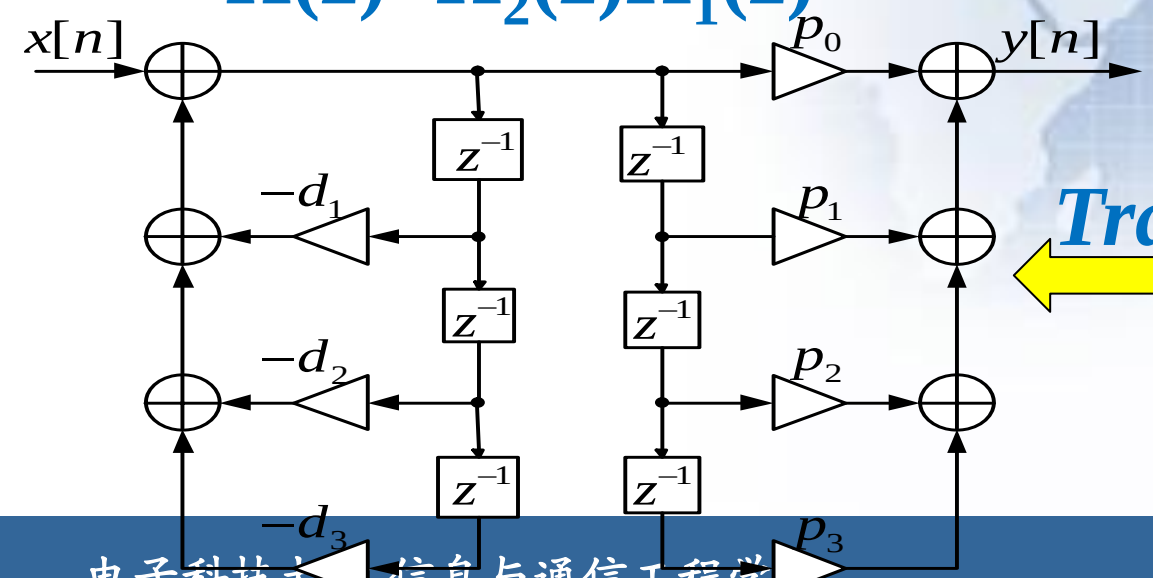
Transpose



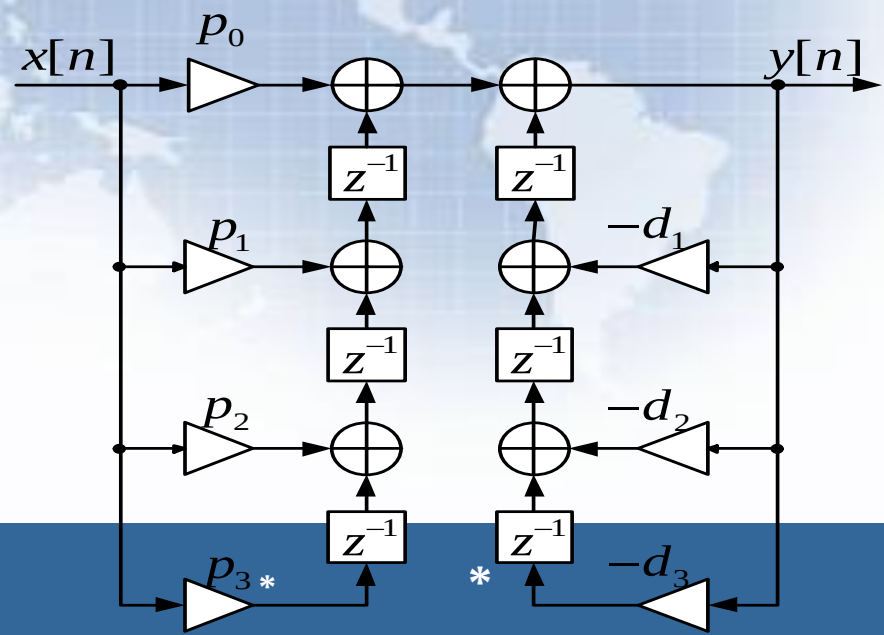


canonic structure

$$H(z) = H_2(z)H_1(z)$$



Transpose

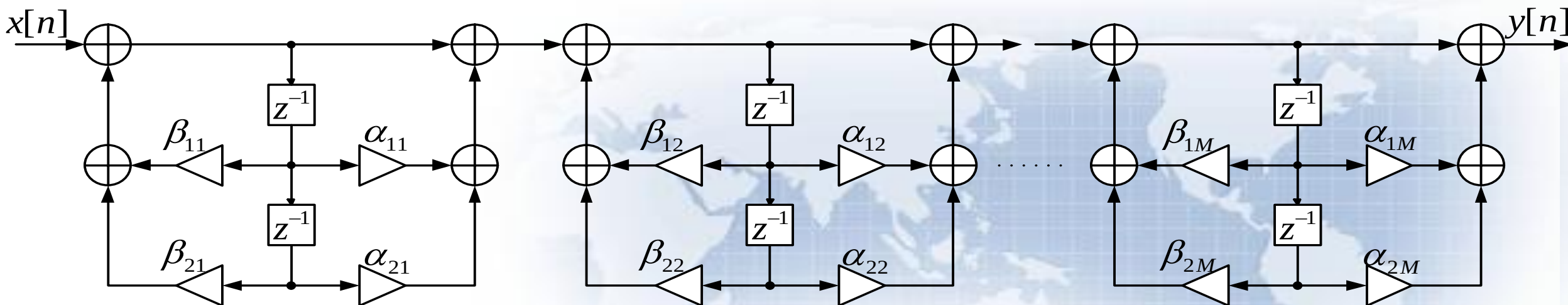


Basic IIR Filter Structure



- Cascade Form**

$$H(z) = H_1(z)H_2(z)...H_M(z)$$



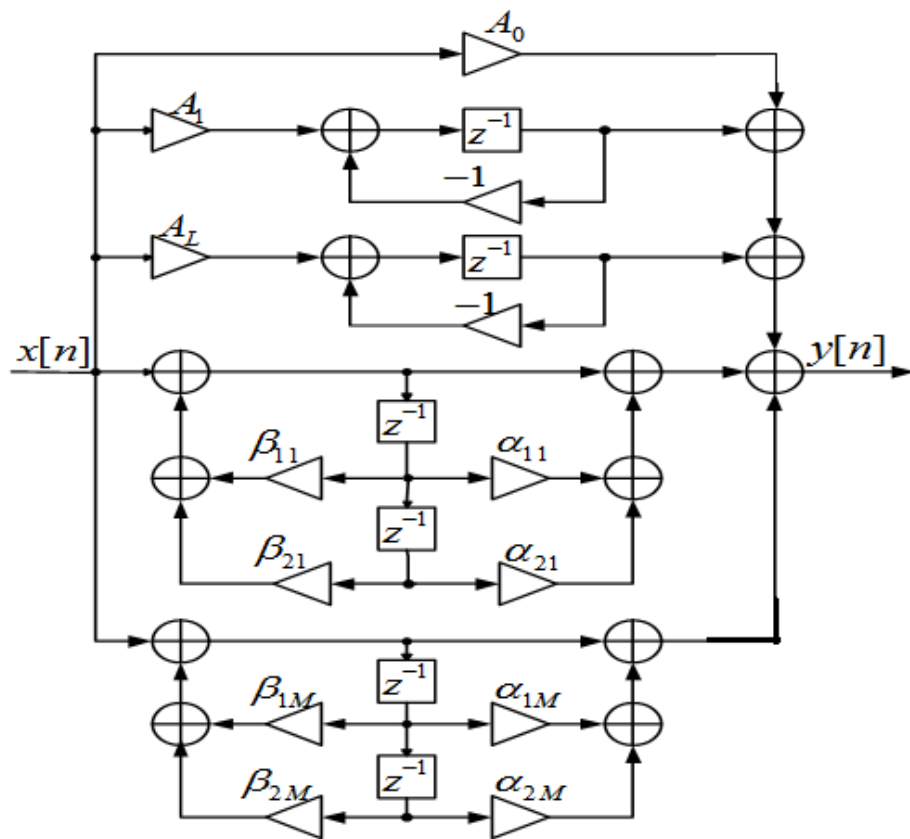
- *It is easy to modify zeros and poles.*
- *But the forms of realization are no identical.*

Basic IIR Filter Structure



- *Parallel Form*

$$H(z) = H_1(z) + H_2(z) + \dots + H_r(z)$$



➤ *It is needed partial-fractional expansion*

➤ *It is easy to modify poles with identical form*

Basic IIR Filter Structure



Example: For a system as below

$$y[n] = x[n] + \frac{1}{3} x[n-1] + \frac{3}{4} y[n-1] - \frac{1}{8} y[n-2]$$

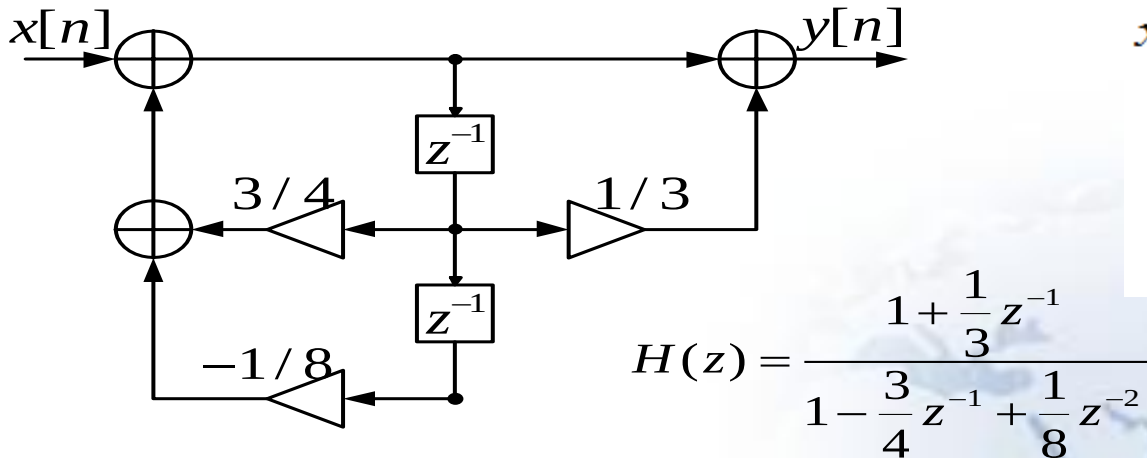
Its transfer function is

$$H(z) = \frac{1 + \frac{1}{3} z^{-1}}{1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}} = \left(\frac{1}{1 - \frac{1}{2} z^{-1}} \right) \left(\frac{1 + \frac{1}{3} z^{-1}}{1 - \frac{1}{4} z^{-1}} \right) = \frac{-7/3}{1 - \frac{1}{4} z^{-1}} + \frac{10/3}{1 - \frac{1}{2} z^{-1}}$$

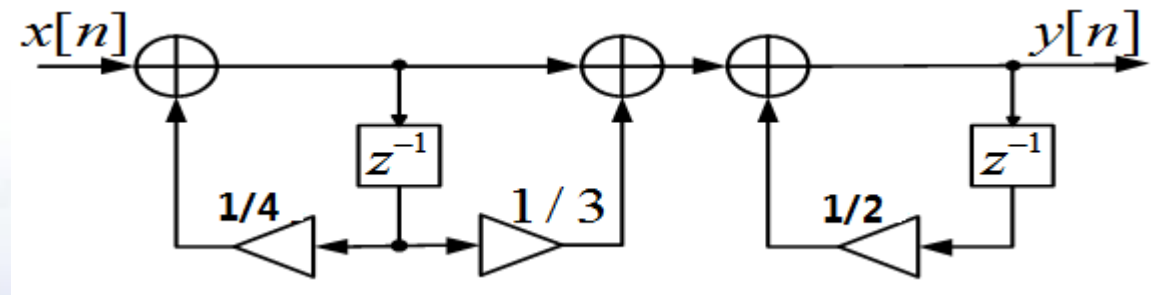
Basic IIR Filter Structure



- Its direct form

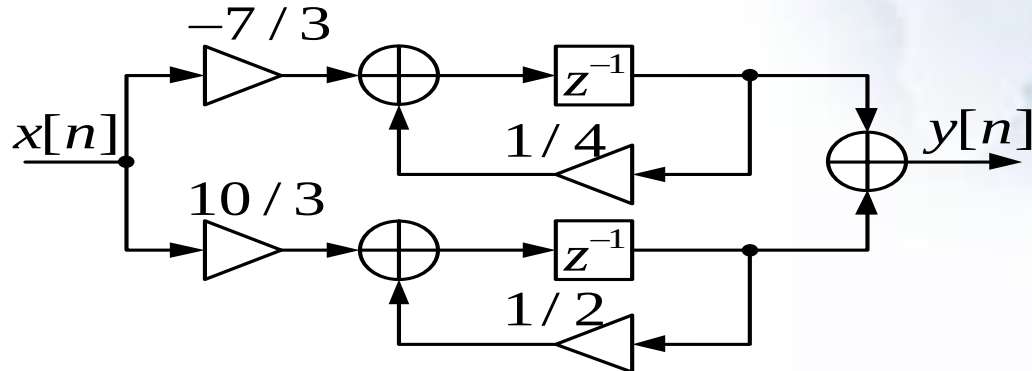


- The cascade form with 1st order



$$H(z) = \left(\frac{1}{1 - \frac{1}{2}z^{-1}} \right) \left(\frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{4}z^{-1}} \right)$$

- The parallel form with first-order



$$H(z) = \frac{-\frac{7}{3}}{1 - \frac{1}{4}z^{-1}} + \frac{\frac{10}{3}}{1 - \frac{1}{2}z^{-1}}$$

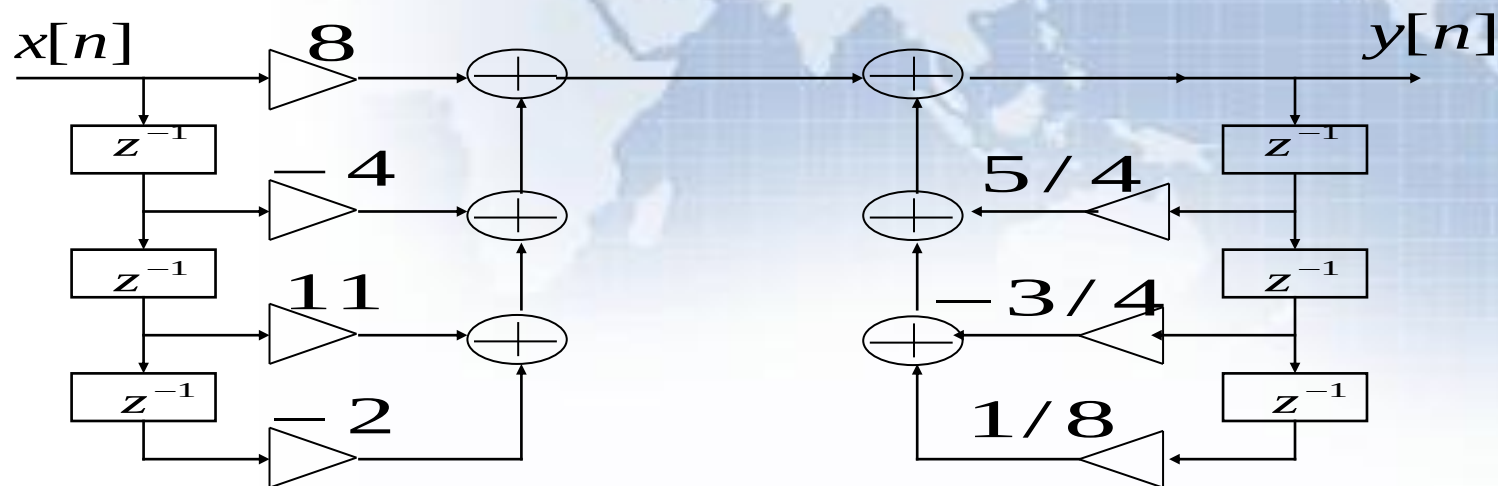
8.4 Basic IIR Structure



Example For a system as below

$$H(z) = \frac{8 - 4z^{-1} + 11z^{-2} - 2z^{-3}}{1 - \frac{5}{4}z^{-1} + \frac{3}{4}z^{-2} - \frac{1}{8}z^{-3}}$$

Its direct form is as below

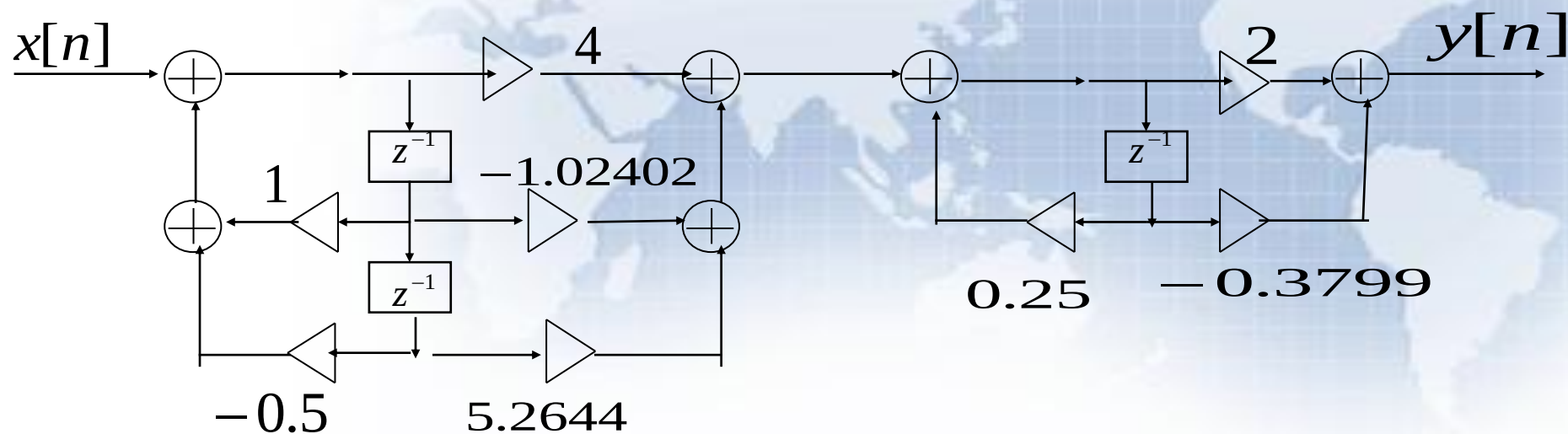


8.4 Basic IIR Structure



Its cascade form with second-order

$$H(z) = \frac{2 - 0.3799z^{-1}}{1 - 0.25z^{-1}} \frac{4 - 1.2402z^{-1} + 5.2644z^{-2}}{1 - z^{-1} + 0.5z^{-2}}$$

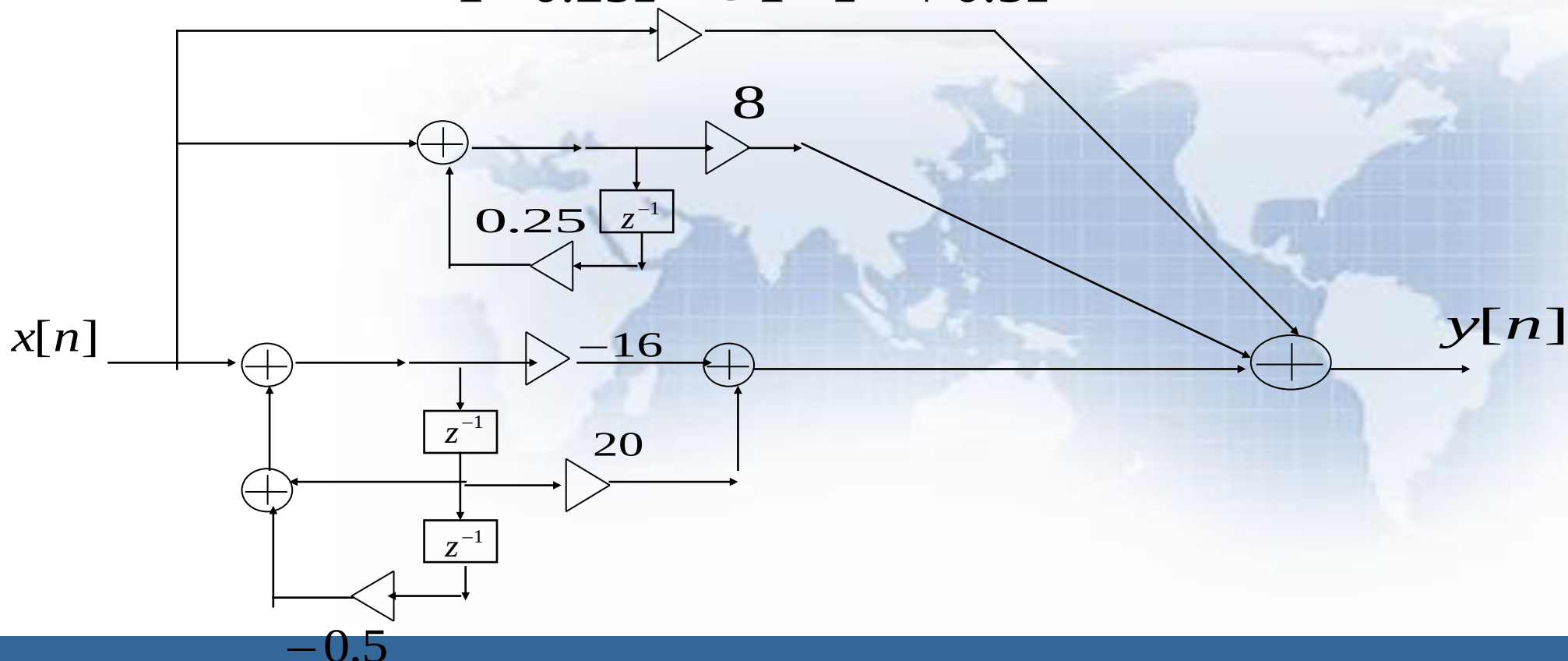


8.4 Basic IIR Structure



Its parallel form with second-order

$$H(z) = 16 + \frac{8}{1 - 0.25z^{-1}} + \frac{-16 + 20z^{-1}}{1 - z^{-1} + 0.5z^{-2}}$$



8.5 Realization Using MATLAB



- The cascade form requires the factorization of the transfer function which can be developed using the M-file **zp2sos**
- The statement **sos = zp2sos(z,p,k)** generates a matrix **sos** containing the coefficients of each 2nd-order section of the equivalent transfer function $H(z)$ determined from its pole-zero form

Homework



- **Problems**

8.5, 8.11, 8.13, 8.25, 8.28

M8.1, M8.2, M8.3